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Sensing the World

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Microcontrollers Reflection

This past week we focused on learning microcontrollers, circuit components, and sensor testing to lay the groundwork for our upcoming final floodgate project. We started by breadboarding a basic circuit with a power supply, potentiometer, and LED. During that setup, we adjusted the potentiometer resistance down under a 20V supply, which immediately blew up the LED. From there, we moved on to switch controls and environmental sensors like the LM393 photoresistor module. We then prototyped a modified version of Dr. Valvano's security system on an Arduino Nano ESP32 using two LEDs, a push button, a potentiometer, and a photoresistor. When our push button kept shorting out and leaving the blue LED permanently lit, we used ChatGPT to figure out the correct 4-pin switch orientation on the breadboard. Later in the week, we practiced controlling a servo motor using PWM signals on pin D5. To finish up, we collected manual voltage data from a raindrop sensor using a Keysight multimeter before writing code to automate 10 ADC voltage readings per wetness condition, saving the output to CSV files, and plotting trendlines in Google Sheets. As a final step, my team (Diego, Diane, and I) filled out our initial planning worksheet and requirements document to outline our big project idea: an automated flood-protection system.

Hands-on circuit building made it clear that getting hardware to behave takes just as much work as writing clean code. Using ChatGPT for code generation was helpful, but we

quickly realized that broad prompts yield useless results unless you specify the exact microchip, GPIO pin numbers, and logic goals upfront. Working through the raindrop sensor data also showed us how easy it is to introduce experimental flaws. Comparing our three manual multimeter checks against our ten automated ADC reads highlighted major variability in our data, mostly because we did not strictly measure or control the amount of moisture on the sensor pad between tests. Overall, getting the servo motor to turn on command and mapping out our initial floodgate concept gave us a solid foundation for building our final project.